📄 AI/ML Engineering Internship Report

Task 3: Heart Disease Prediction Using Machine Learning

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1. Introduction

Heart disease is one of the leading causes of death worldwide. Early prediction of cardiovascular diseases can significantly help healthcare professionals take preventive measures and reduce patient risk. Machine Learning techniques provide intelligent solutions by analyzing historical medical data and identifying hidden patterns.

The objective of this project is to develop a **binary classification model** capable of predicting whether a person is at risk of heart disease based on clinical and health-related attributes. This task was completed as part of the **AI/ML Engineering Internship at DevelopersHub Corporation**, focusing on practical implementation of Artificial Intelligence concepts in healthcare analytics.

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2. Problem Statement

Traditional medical diagnosis requires extensive testing and expert evaluation, which can be time-consuming and costly. The goal of this project is to build an automated prediction system that assists in identifying heart disease risk using patient health data.

The system analyzes multiple medical parameters such as age, cholesterol level, blood pressure, and heart rate to predict the likelihood of heart disease.

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3. Dataset Description

The project uses the **Heart Disease UCI Dataset**, publicly available on Kaggle. The dataset contains patient medical records collected for cardiovascular research.

Key Attributes:

Feature	Description
Age	Patient age
Sex	Gender
Chest Pain Type	Type of chest discomfort
Resting Blood Pressure	Blood pressure level
Cholesterol	Serum cholesterol level
Maximum Heart Rate	Achieved heart rate
Oldpeak	ST depression measurement
Target	Heart disease presence (1 = Yes, 0 = No)

The dataset represents a **supervised learning problem** where the target variable indicates disease presence.

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4. Data Preprocessing

Data preprocessing is an essential step in machine learning pipelines to ensure model reliability.

The following preprocessing steps were performed:

- Dataset loading using Pandas.
- Identification of missing values.
- Replacement of missing values using mean imputation.
- Verification of data types and consistency.
- Separation of independent variables (features) and dependent variable (target).

Proper preprocessing improved model learning efficiency and minimized bias.

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5. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to understand relationships between medical features and heart disease occurrence.

EDA techniques included:

- Distribution analysis of heart disease cases.
- Correlation heatmap visualization.
- Feature relationship inspection.

The analysis revealed that chest pain type, maximum heart rate, cholesterol levels, and age show strong correlation with heart disease prediction.

EDA helps transform raw medical data into meaningful insights before model training.

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6. Model Development

Since the task involves predicting two possible outcomes (disease or no disease), it is categorized as a **binary classification problem**.

Selected Machine Learning Models:

1. Logistic Regression

Logistic Regression was used as the primary model due to its effectiveness in medical classification problems and interpretability.

2. Decision Tree Classifier

A Decision Tree model was also implemented for comparison purposes.

The dataset was divided into:

- Training Set: 80%
- Testing Set: 20%

This ensures unbiased performance evaluation.

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7. Model Evaluation

Model performance was evaluated using standard AI/ML metrics.

Accuracy

Accuracy measures the proportion of correctly predicted instances.

The Logistic Regression model achieved approximately **85–90% accuracy**, indicating strong predictive capability.

Confusion Matrix

The confusion matrix evaluates prediction correctness by comparing actual and predicted values.

It identifies:

- True Positives
- True Negatives
- False Positives
- False Negatives

This metric is particularly important in healthcare applications where incorrect predictions may affect patient outcomes.

ROC Curve and AUC Score

The Receiver Operating Characteristic (ROC) curve evaluates model discrimination ability across classification thresholds.

The model achieved an **ROC-AUC score greater than 0.80**, demonstrating reliable classification performance.

A higher AUC value indicates better separation between healthy and at-risk patients.

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8. Feature Importance Analysis

Feature importance analysis was conducted to identify medical attributes influencing prediction results.

The most influential factors included:

- Chest Pain Type
- Maximum Heart Rate
- Cholesterol Level
- Age
- ST Depression (Oldpeak)

Understanding feature importance improves transparency and trust in AI-based healthcare systems.

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9. Results and Discussion

The developed machine learning model successfully predicts heart disease risk using patient health indicators. Logistic Regression provided stable and interpretable results suitable for medical applications.

The project demonstrates how Artificial Intelligence can assist healthcare professionals by providing decision-support systems rather than replacing medical expertise.

Machine learning models like this can be integrated into hospital management systems or health monitoring applications for early disease detection.

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10. Challenges Faced

During implementation, several challenges were encountered:

- Understanding medical dataset attributes.
- Handling missing values effectively.
- Selecting appropriate evaluation metrics.
- Preventing model overfitting.

These challenges were resolved through preprocessing techniques and proper model validation.

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11. Skills Gained

Through this task, the following AI/ML engineering skills were developed:

- Data preprocessing and cleaning
- Exploratory Data Analysis
- Binary classification modeling
- Model evaluation techniques
- ROC-AUC interpretation
- Feature importance analysis
- Medical data interpretation

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12. Conclusion

This project successfully implemented a Heart Disease Prediction System using Machine Learning techniques. The developed model demonstrates high accuracy and reliable performance in identifying individuals at risk of heart disease.

The project highlights the practical application of Artificial Intelligence in healthcare analytics and emphasizes the importance of data-driven decision-making systems.

Future improvements may include deep learning models, real-time patient monitoring systems, and deployment using web-based applications.

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13. Future Work

Potential future enhancements include:

- Deployment using Streamlit or Flask.
- Integration with hospital databases.
- Real-time prediction dashboards.
- Deep learning-based prediction models.
- Mobile healthcare applications.

✅ **Internship Level:** AI/ML Engineering
✅ **Domain:** Healthcare Analytics
✅ **Model Type:** Binary Classification
✅ **Tools Used:** Python, Pandas, Scikit-learn, Matplotlib, Seaborn

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